

Finite-Volume Diagonal-Split Solvers for a 2D Fokker–Planck

Notes for Call Nr. 5

Cédric Zeiter

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Colorcode meaning:

Repetition in Orange (mostly unchanged, can be ignored)

Revised method (finit volume, $\Delta x \neq \Delta y$) in black

First calculations for more realistic scheme in Purple

Very brief thoughts about source term in Olive

1 PDE and Physical Setup

$$\frac{\partial C}{\partial t} = -\frac{\partial(F_x C)}{\partial x} - \frac{\partial(F_y C)}{\partial y} + \frac{\partial^2(D_{xx} C)}{\partial x^2} + \frac{\partial^2(D_{yy} C)}{\partial y^2} + \frac{\partial^2(D_{xy} C)}{\partial x \partial y}. \quad (1)$$

$$\mathcal{M} = \{(0, 1), (1, 1)\}. \quad (2)$$

$$d(x) = \begin{cases} \max(\beta(x)c_m + \alpha(x), 0), & x \geq x_{\text{activation}}, \\ 0, & x < x_{\text{activation}}, \end{cases} \quad (3)$$

with

$$\beta(x) = \sqrt{x}, \quad \alpha(x) = \beta(x)e^{-E(x)}, \quad E(x) = 71.29 - 114.81 \left(x^{2/3} - (x-1)^{2/3} \right). \quad (4)$$

1.1 Explicit tensor sums over $\mathcal{M} = \{(0, 1), (1, 1)\}$

Let (m, q) denote coordinates of each mobile direction. For $\mathcal{M} = \{(0, 1), (1, 1)\}$ we have $m \in \{0, 1\}$ and $q \in \{1, 1\}$, with the same scalar weight d .

Step 1: D_{xx}

$$D_{xx} = \frac{1}{2} \sum_{(m,q) \in \mathcal{M}} m^2 d = \frac{1}{2} ((0)^2 d + (1)^2 d) = \frac{d}{2}. \quad (5)$$

Step 2: D_{yy}

$$D_{yy} = \frac{1}{2} \sum_{(m,q) \in \mathcal{M}} q^2 d = \frac{1}{2} ((1)^2 d + (1)^2 d) = d. \quad (6)$$

Step 3: D_{xy}

$$D_{xy} = \sum_{(m,q) \in \mathcal{M}} m q d = (0 \cdot 1) d + (1 \cdot 1) d = d. \quad (7)$$

Therefore

$$D_{xx} = \frac{d}{2}, \quad D_{yy} = d, \quad D_{xy} = d. \quad (8)$$

1.2 Analytical operator identity

The diffusion part of (1) reads

$$\text{Diff}(C) = \partial_{xx}(D_{xx}C) + \partial_{yy}(D_{yy}C) + \partial_{xy}(D_{xy}C). \quad (9)$$

Introduce the half cross-coefficient and residual axial coefficients:

$$D_2 := \frac{D_{xy}}{2} = \frac{d}{2}, \quad A_x := D_{xx} - D_2 = 0, \quad A_y := D_{yy} - D_2 = \frac{d}{2}. \quad (10)$$

Using the identity

$$(\partial_x + \partial_y)^2 = \partial_{xx} + 2\partial_{xy} + \partial_{yy} \quad (11)$$

one verifies:

$$\partial_{xx}(A_x C) + \partial_{yy}(A_y C) + (\partial_x + \partial_y)^2(D_2 C) = \partial_{xx}\left(\frac{d}{2}C\right) + \partial_{yy}(dC) + \partial_{xy}(dC) = \text{Diff}(C). \quad (12)$$

Since $A_x = 0$ this simplifies to

$$\text{Diff}(C) = \partial_{yy}(A_y C) + (\partial_x + \partial_y)^2(D_2 C). \quad (13)$$

Equation (13) unfortunately doesn't hold anymore, when we redefine the mesh, as (11) doesn't hold anymore.

1.3 Diagonal finite-difference stencil for general $\Delta x, \Delta y$

In this subsection, U denotes a generic scalar field sampled on the grid. In the diffusion split below we apply the same stencil either to $U = C$ (constant coefficients) or, locally, to coefficient-weighted quantities such as $U = \tilde{D}_2 C$.

A symmetric Taylor expansion about (x_i, y_j) gives:

$$U_{i+1,j+1} = U_{i,j} + \Delta x \partial_x U + \Delta y \partial_y U + \frac{\Delta x^2}{2} \partial_{xx} U + \Delta x \Delta y \partial_{xy} U + \frac{\Delta y^2}{2} \partial_{yy} U + \mathcal{O}(h^3), \quad (14)$$

$$U_{i-1,j-1} = U_{i,j} - \Delta x \partial_x U - \Delta y \partial_y U + \frac{\Delta x^2}{2} \partial_{xx} U + \Delta x \Delta y \partial_{xy} U + \frac{\Delta y^2}{2} \partial_{yy} U + \mathcal{O}(h^3), \quad (15)$$

where $h = \max(\Delta x, \Delta y)$. Adding the two expansions¹ and rearranging gives:

$$U_{i+1,j+1} + U_{i-1,j-1} - 2U_{i,j} = \Delta x^2 \partial_{xx} U + 2\Delta x \Delta y \partial_{xy} U + \Delta y^2 \partial_{yy} U + \mathcal{O}(h^4). \quad (16)$$

Dividing through by $\Delta x \Delta y$:

$$\boxed{\frac{U_{i+1,j+1} - 2U_{i,j} + U_{i-1,j-1}}{\Delta x \Delta y} = \frac{\Delta x}{\Delta y} \partial_{xx} U + 2 \partial_{xy} U + \frac{\Delta y}{\Delta x} \partial_{yy} U + \mathcal{O}(h^2).} \quad (17)$$

$$\mathcal{T}_h \phi := r \partial_{xx} \phi + 2 \partial_{xy} \phi + \frac{1}{r} \partial_{yy} \phi, \quad r := \frac{\Delta x}{\Delta y}. \quad (18)$$

Key observations

- When $\Delta x = \Delta y$ (i.e. $r = 1$), $\mathcal{T}_h = \partial_{xx} + 2\partial_{xy} + \partial_{yy} = (\partial_x + \partial_y)^2$, recovering (11).
- For $r \neq 1$, the analytical split (13) must be redefined $\Rightarrow$ need new splitting operators.

¹Note that the $\mathcal{O}(h)$ have opposite signs in the two formulas and that they cancel. Only even $\mathcal{O}(h^x)$ stay

1.4 Mesh-adapted operator splitting

We seek $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ such that

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C), \quad (19)$$

After $\mathcal{T}_h$ via (18) and reorder the derivatives, we want to find (9) again:

$$\partial_{xx}(\tilde{A}_x + r \tilde{D}_2)C + \partial_{yy}(\tilde{A}_y + \frac{1}{r} \tilde{D}_2)C + 2\tilde{D}_2 \partial_{xy}C = \partial_{xx}(D_{xx}C) + \partial_{yy}(D_{yy}C) + \partial_{xy}(D_{xy}C). \quad (20)$$

Now match coefficients of $\partial_{xx}, \partial_{yy}, \partial_{xy}$:

$$\partial_{xy} : \quad 2\tilde{D}_2 = D_{xy} \implies \tilde{D}_2 = \frac{D_{xy}}{2}, \quad (21)$$

$$\partial_{xx} : \quad \tilde{A}_x + r \tilde{D}_2 = D_{xx} \implies \tilde{A}_x = D_{xx} - r \tilde{D}_2, \quad (22)$$

$$\partial_{yy} : \quad \tilde{A}_y + \frac{1}{r} \tilde{D}_2 = D_{yy} \implies \tilde{A}_y = D_{yy} - \frac{\tilde{D}_2}{r}. \quad (23)$$

Substituting the original mesh values we had (8) and insert $r = \Delta x / \Delta y$:

Step 1: $\tilde{D}_2$

$$\tilde{D}_2 = \frac{D_{xy}}{2} = \frac{d}{2}. \quad (24)$$

The half cross-coefficient is independent of the grid.

Step 2: $\tilde{A}_x$

$$\tilde{A}_x = D_{xx} - r \tilde{D}_2 = \frac{d}{2} - \frac{\Delta x}{\Delta y} \cdot \frac{d}{2} = \frac{d}{2} \left(1 - \frac{\Delta x}{\Delta y}\right). \quad (25)$$

When $\Delta x = \Delta y$: $\tilde{A}_x = 0$, recovering $A_x = 0$ from (10). **When $\Delta x < \Delta y$: $\tilde{A}_x > 0$, so an additional x-axial diffusion flux must be included.**

Step 3: $\tilde{A}_y$

$$\tilde{A}_y = D_{yy} - \frac{\tilde{D}_2}{r} = d - \frac{d/2}{\Delta x / \Delta y} = d - \frac{d \Delta y}{2 \Delta x} = d \left(1 - \frac{\Delta y}{2 \Delta x}\right) = \frac{d(2 \Delta x - \Delta y)}{2 \Delta x}. \quad (26)$$

When $\Delta x = \Delta y$: $\tilde{A}_y = d/2$, recovering $A_y = d/2$ from (10).

2 Finite Volume Formulation

We work on a Cartesian grid with uniform spacings Δx (in x) and Δy (in y). Cell (i, j) occupies

$$\Omega_{i,j} = [x_{i-1/2}, x_{i+1/2}] \times [y_{j-1/2}, y_{j+1/2}], \quad |\Omega_{i,j}| = \Delta x \Delta y. \quad (27)$$

The cell average is

$$U_{i,j}(t) \approx \frac{1}{\Delta x \Delta y} \int_{\Omega_{i,j}} C(x, y, t) dx dy. \quad (28)$$

Integrating (1) over $\Omega_{i,j}$ and applying the divergence theorem gives

$$\frac{dU_{i,j}}{dt} = \text{adv}_{i,j} + \text{diff}_{i,j}, \quad (29)$$

with the right-hand side expressed as conservative flux differences derived below.

3 Mass Conservation and Flux-Form Discretisation

To enforce exact mass conservation the diffusion operator must be cast entirely in flux form:

$$\partial_t U + \nabla \cdot \mathbf{J} = 0, \quad \mathbf{J} = \mathbf{J}_{\text{adv}} + \mathbf{J}_{\text{diff}}. \quad (30)$$

Each face flux is computed once and applied with opposite signs in the two cells sharing that face, so internal contributions cancel in the global mass sum.

3.1 FV advection fluxes (upwind, conservative) — Kept same as last week

Interface drift speeds (arithmetic cell-average):

$$v_{i+\frac{1}{2},j}^x = \frac{1}{2} (F_{x,i,j} + F_{x,i+1,j}), \quad (31)$$

$$v_{i,j+\frac{1}{2}}^y = \frac{1}{2} (F_{y,i,j} + F_{y,i,j+1}), \quad (32)$$

Upwind fluxes in x and y direction:

$$F_{i+\frac{1}{2},j}^{\text{adv}} = \begin{cases} v_{i+\frac{1}{2},j}^x u_{i,j}^n, & v_{i+\frac{1}{2},j}^x \geq 0, \\ v_{i+\frac{1}{2},j}^x u_{i+1,j}^n, & v_{i+\frac{1}{2},j}^x < 0, \end{cases} \quad (33)$$

$$F_{i,j+\frac{1}{2}}^{\text{adv}} = \begin{cases} v_{i,j+\frac{1}{2}}^y u_{i,j}^n, & v_{i,j+\frac{1}{2}}^y \geq 0, \\ v_{i,j+\frac{1}{2}}^y u_{i,j+1}^n, & v_{i,j+\frac{1}{2}}^y < 0, \end{cases} \quad (34)$$

and contributes through discrete divergence:

$$\text{adv}_{i,j} = -\frac{F_{i+\frac{1}{2},j}^{\text{adv}} - F_{i-\frac{1}{2},j}^{\text{adv}}}{h} - \frac{F_{i,j+\frac{1}{2}}^{\text{adv}} - F_{i,j-\frac{1}{2}}^{\text{adv}}}{h}. \quad (35)$$

This is basically a linear reduction of Godunov, ensuring that the same amount of mass enters and leaves grid-cells.

3.2 FV diffusion fluxes

From (19) we have:

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C), \quad (36)$$

3.2.1 $\tilde{A}_x$ part

Write $\partial_{xx}(\tilde{A}_x C) = \partial_x G^x$ with

$$G^x = \partial_x(\tilde{A}_x C). \quad (37)$$

Face approximation at $(i + 1/2, j)$ with arithmetic coefficient averaging:

$$G_{i+1/2,j}^x = \tilde{A}_{x,i+1/2,j} \frac{U_{i+1,j} - U_{i,j}}{\Delta x}, \quad \tilde{A}_{x,i+1/2,j} = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i+1,j}}{2}. \quad (38)$$

Cell divergence:

$$\text{diff}_{i,j}^{(x)} = \frac{G_{i+1/2,j}^x - G_{i-1/2,j}^x}{\Delta x}. \quad (39)$$

Boundary fluxes are set to zero: $G_{1/2,j}^x = G_{N_x+1/2,j}^x = 0$.

3.2.2 $\tilde{A}_y$ part

Identical idea as in 3.2.1

Write $\partial_{yy}(\tilde{A}_y C) = \partial_y G^y$ with $G^y = \partial_y(\tilde{A}_y C)$. Face approximation at $(i, j + 1/2)$:

$$G_{i,j+1/2}^y = \tilde{A}_{y,i,j+1/2} \frac{U_{i,j+1} - U_{i,j}}{\Delta y}, \quad \tilde{A}_{y,i,j+1/2} = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j+1}}{2}. \quad (40)$$

Cell divergence:

$$\text{diff}_{i,j}^{(y)} = \frac{G_{i,j+1/2}^y - G_{i,j-1/2}^y}{\Delta y} \quad (41)$$

Boundary fluxes: $G_{i,1/2}^y = G_{i,N_y+1/2}^y = 0$.

3.2.3 Diagonal part.

The stencil (17) motivates the following corner flux (Treating diagonal like a usual finite difference approximation, but with Δx and Δy in denominator). At the north-east corner $(i + 1/2, j + 1/2)$, define

$$G_{i+1/2,j+1/2}^d = \tilde{D}_{2,i+1/2,j+1/2} (U_{i+1,j+1} - U_{i,j}), \quad \tilde{D}_{2,i+1/2,j+1/2} = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i+1,j+1}}{2}. \quad (42)$$

$$\text{diff}_{i,j}^{(d)} = \frac{G_{i+1/2,j+1/2}^d - G_{i-1/2,j-1/2}^d}{\Delta x \Delta y}. \quad (43)$$

Boundary corners with no physical neighbour are assigned $G^d = 0$ (zero-flux / reflecting condition along the diagonal direction).

Consistency with the diagonal stencil. For constant $\tilde{D}_2$:

$$\begin{aligned} \text{diff}_{i,j}^{(d)} &= \frac{1}{\Delta x \Delta y} \left[\tilde{D}_2 (U_{i+1,j+1} - U_{i,j}) - \tilde{D}_2 (U_{i,j} - U_{i-1,j-1}) \right] \\ &= \tilde{D}_2 \frac{U_{i+1,j+1} - 2U_{i,j} + U_{i-1,j-1}}{\Delta x \Delta y}, \end{aligned} \quad (44)$$

which by (17) consistently approximates $\tilde{D}_2 \mathcal{T}_h(U) = \tilde{D}_2 (r \partial_{xx} + 2 \partial_{xy} + r^{-1} \partial_{yy}) U$.

4 Implicit Diffusion Matrix Assembly (IMEX)

Time splitting:

$$U^* = U^n + \Delta t \text{adv}(U^n), \quad (45)$$

$$(I - \Delta t L_{\text{diff}}) U^{n+1} = U^*. \quad (46)$$

Row-major index: $k = (i - 1)N_y + j$.

4.1 Interior row stencil

For interior (i, j) we define six face-averaged diffusion coefficients (reason on why they are defined like this follows shortly)

$$\tilde{a}_{i+1/2}^x = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i+1,j}}{2 \Delta x^2}, \quad \tilde{a}_{i-1/2}^x = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i-1,j}}{2 \Delta x^2}, \quad (47)$$

$$\tilde{a}_{j+1/2}^y = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j+1}}{2 \Delta y^2}, \quad \tilde{a}_{j-1/2}^y = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j-1}}{2 \Delta y^2}, \quad (48)$$

$$\tilde{d}_+ = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i+1,j+1}}{2 \Delta x \Delta y}, \quad \tilde{d}_- = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i-1,j-1}}{2 \Delta x \Delta y}. \quad (49)$$

Now we look at all the cells affected around a given cell. This is captured in L.

$$L_{k,k} += -(\tilde{a}_{i+1/2}^x + \tilde{a}_{i-1/2}^x + \tilde{a}_{j+1/2}^y + \tilde{a}_{j-1/2}^y + \tilde{d}_+ + \tilde{d}_-), \quad (50)$$

$$L_{k,k(i+1,j)} += \tilde{a}_{i+1/2}^x, \quad L_{k,k(i-1,j)} += \tilde{a}_{i-1/2}^x, \quad (51)$$

$$L_{k,k(i,j+1)} += \tilde{a}_{j+1/2}^y, \quad L_{k,k(i,j-1)} += \tilde{a}_{j-1/2}^y, \quad (52)$$

$$L_{k,k(i+1,j+1)} += \tilde{d}_+, \quad L_{k,k(i-1,j-1)} += \tilde{d}_-. \quad (53)$$

When inserting the upper equations into the L equations we can recover exactly diff-notations (39), (41), and (43) (when looking at what cells are impacted in diff formulation).

4.2 Diagonal dominance of $A = I - \Delta t L$

Non-negativity and mesh-aspect-ratio constraint. Provided the mesh satisfies (60), all coefficients $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2 \geq 0$, which ensures:

$$L_{kk} \leq 0, \quad L_{k\ell} \geq 0 \ (\ell \neq k), \quad \sum_{\ell} L_{k\ell} = 0. \quad (54)$$

For $A = I - \Delta t L$:

$$A_{kk} = 1 + \Delta t |L_{kk}|, \quad A_{k\ell} = -\Delta t L_{k\ell} \leq 0 \ (\ell \neq k),^2 \quad (55)$$

and

$$A_{kk} - \sum_{\ell \neq k} |A_{k\ell}| = 1 + \Delta t |L_{kk}| - \Delta t \sum_{\ell \neq k} L_{k\ell} = 1 + \Delta t (|L_{kk}| - \sum_{\ell \neq k} L_{k\ell}) = 1 + \Delta t (\sum_{\ell} L_{k\ell}) = 1 > 0. \quad (56)$$

Hence A is strictly diagonally dominant by (hence non-singular), guaranteeing a unique solution to the linear system.

Constraint introduced For what we just did to find the unique solution we required:

$$\tilde{A}_x \geq 0 \iff 1 - \frac{\Delta x}{\Delta y} \geq 0 \iff \Delta x \leq \Delta y, \quad (57)$$

$$\tilde{A}_y \geq 0 \iff 2 \Delta x - \Delta y \geq 0 \iff \Delta y \leq 2 \Delta x. \quad (58)$$

For the diagonal part we also need

$$\tilde{D}_2 \geq 0 \iff \frac{D_{xy}}{2} \geq 0 \iff D_{xy} \geq 0. \quad (59)$$

²Just for me as reminder: $I - dt * L$ is $-dt * L$ as I only has elements on diagonal.

In our reduced model $D_{xy} = d \geq 0$, hence $\tilde{D}_2 = d/2 \geq 0$ automatically.

A more physical explanation on why these operator have to be positive is that we would otherwise have anti-diffusion.

Both axial conditions hold simultaneously when

$$\boxed{\frac{1}{2} \leq r \leq 1, \quad r := \frac{\Delta x}{\Delta y}.} \quad (60)$$

Outside this band, one axial coefficient becomes negative, potentially destabilising the scheme.

5 CFL and Stability

5.1 Explicit FV method

Advection bound. With separate spacings, the multi-dimensional upwind CFL reads ³:

$$\Delta t \left(\frac{\|F_x\|_\infty}{\Delta x} + \frac{\|F_y\|_\infty}{\Delta y} \right) \leq 1. \quad (61)$$

Diffusion bound. From standard stability analysis follows:

$$\Delta t \frac{2\|\tilde{A}_x\|_\infty}{\Delta x^2} \leq 1, \quad \Delta t \frac{2\|\tilde{A}_y\|_\infty}{\Delta y^2} \leq 1. \quad (62)$$

For the diagonal part, this is not so standard anymore. I decided to just assume the same bound as for the other directions, which intuitively makes sense to me. But this needs to be checked mathematically.

$$\Delta t \frac{2\|\tilde{D}_2\|_\infty}{\Delta x \Delta y} \leq 1. \quad (63)$$

After making all bounds tight and adding up the terms (by triangular inequality bound still holds):

$$\Delta t_{\text{diff}} = \frac{1}{2} \left(\frac{\|\tilde{A}_x\|_\infty}{\Delta x^2} + \frac{\|\tilde{A}_y\|_\infty}{\Delta y^2} + \frac{\|\tilde{D}_2\|_\infty}{\Delta x \Delta y} \right)^{-1}. \quad (64)$$

The overall time step:

$$\Delta t = \min \left(\frac{1}{\|F_x\|_\infty/\Delta x + \|F_y\|_\infty/\Delta y}, \Delta t_{\text{diff}} \right). \quad (65)$$

5.2 IMEX method

With implicit diffusion, the parabolic stiffness is resolved by solving

$$(I - \Delta t L_{\text{diff}}) U^{n+1} = U^*. \quad (66)$$

The parabolic restrictions Δt_{diff} are removed from time-step selection. Only the hyperbolic advection CFL:

$$\Delta t \left(\frac{\|F_x\|_\infty}{\Delta x} + \frac{\|F_y\|_\infty}{\Delta y} \right) \leq 1 \quad (67)$$

must be satisfied.

Note that for the derivation to work also a requirement on the grid must hold (as shown above):

$$\boxed{\frac{1}{2} \leq r \leq 1, \quad r := \frac{\Delta x}{\Delta y}.} \quad (68)$$

³Stricter bound than in literature like [1]

5.3 Global mass and roundoff correction.

In numerical experiments slight drift in the order of 10^{-13} were observed, to stabilize the method even further these small drifts get removed in our method.

The total mass is monitored at each step:

$$M^n = \Delta x \Delta y \sum_{i,j} U_{i,j}^n. \quad (69)$$

With threshold $\tau_{\text{round}} = 10^{-10}$,

$$\delta_n = \frac{|M^n - M^0|}{\max(|M^0|, 10^{-30})}, \quad \delta_n \leq \tau_{\text{round}} \implies U^n \leftarrow U^n \frac{M^0}{M^n}. \quad (70)$$

Only machine-precision drift is removed. Fig. 1 shows the mass drift in the mass-corrected IMEX finite volume method. Fig. 2 shows the IMEX method without correction. The fully-explicit method shown in Fig. 3 shows even smaller mass drift. **It is yet unclear to me if this kind of correction makes our model unphysical.**

Figure 1: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method (purely explicit method looks almost identical). Uses mass-correction.

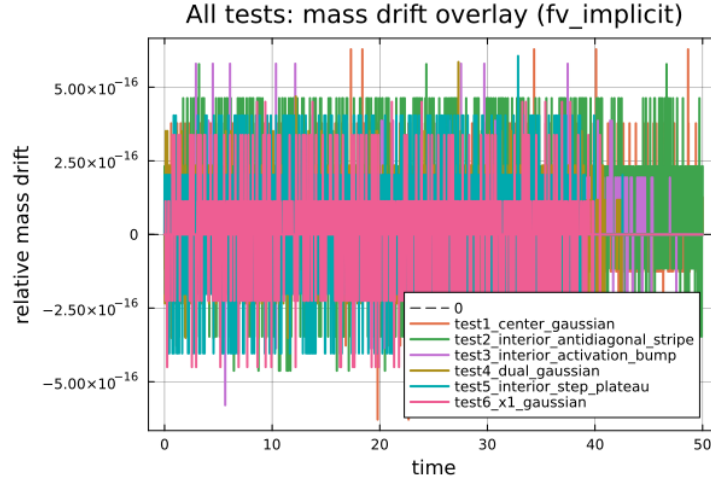


Figure 2: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method. Does **not** use mass-correction.

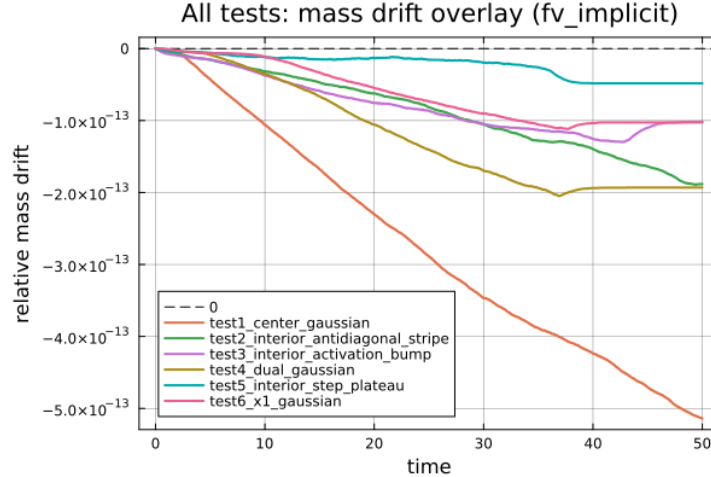
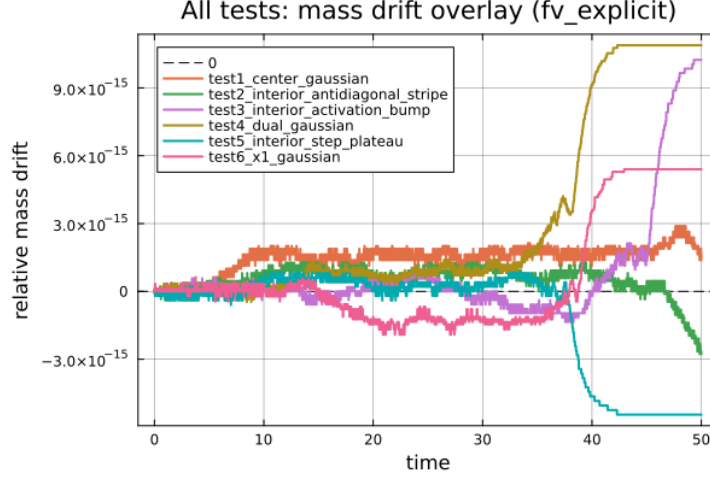


Figure 3: Relative total mass drift on entire domain over time for six different tests with Finite Volume explicit method. Does **not** use mass-correction.



6 Extended Model: $\mathcal{M} = \{(1, 0), (1, 1)\}$ with Two Kinetic Rates

6.1 Physical setup and kinetic fields

Now uses two mobile species:

$$\mathcal{M} = \{(1, 0), (1, 1)\}. \quad (71)$$

Their beta-rates are now independent, parametrised by $\epsilon > 0$:

$$\beta_{(1,0)}(x) = x^{1/2}, \quad \beta_{(1,1)}(x) = \epsilon x^{1/2}, \quad (72)$$

$$\alpha_{(1,0)}(x) = \beta_{(1,0)} e^{-E_{(1,0)}(x)}, \quad \alpha_{(1,1)}(x) = \beta_{(1,1)} e^{-E_{(1,1)}(x)}, \quad (73)$$

with energies (lvalid for $x \geq 2$)

$$E_{(1,0)}(x) = 24.47 - 34(x^{2/3} - (x-1)^{2/3}) \quad (74)$$

$$E_{(1,1)}(x) = 27.47 - 34(x^{2/3} - (x-1)^{2/3}). \quad (75)$$

Introduce two independent scalar kinetic fields:

$$d_1(x) = \beta_{(1,0)}(x) C_{(1,0)} + \alpha_{(1,0)}(x), \quad (76)$$

$$d_2(x) = \beta_{(1,1)}(x) C_{(1,1)} + \alpha_{(1,1)}(x), \quad (77)$$

where $C_{(1,0)}, C_{(1,1)} \geq 0$ are the mobile-species concentrations.

Both fields are non-negative by construction.

6.2 Explicit tensor sums over $\mathcal{M} = \{(1, 0), (1, 1)\}$

Apply to the formulas for the different diffusions.

$$D_{xx} = \frac{1}{2} \sum_{(m,q) \in \mathcal{M}} m^2 d_{(m,q)} = \frac{1}{2} (1^2 \cdot d_1 + 1^2 \cdot d_2) = \frac{d_1 + d_2}{2}. \quad (78)$$

$$D_{yy} = \frac{1}{2} \sum_{(m,q) \in \mathcal{M}} q^2 d_{(m,q)} = \frac{1}{2} (0^2 \cdot d_1 + 1^2 \cdot d_2) = \frac{d_2}{2}. \quad (79)$$

$$D_{xy} = \sum_{(m,q) \in \mathcal{M}} m q d_{(m,q)} = (1 \cdot 0) d_1 + (1 \cdot 1) d_2 = d_2. \quad (80)$$

Therefore

$$\boxed{D_{xx} = \frac{d_1 + d_2}{2}, \quad D_{yy} = \frac{d_2}{2}, \quad D_{xy} = d_2.} \quad (81)$$

Note:

$$D_{xy} = 2 D_{yy}, \quad (82)$$

6.3 Operators for general $r = \frac{\Delta x}{\Delta y}$

The diffusion part of (1) with (81) reads

$$\text{Diff}(C) = \partial_{xx} \left(\frac{d_1 + d_2}{2} C \right) + \partial_{yy} \left(\frac{d_2}{2} C \right) + \partial_{xy} (d_2 C). \quad (83)$$

The diagonal stencil operator $\mathcal{T}_h$ defined in (18) and the general splitting idea from before stays unchanged.

We seek $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ satisfying (19) and matching (83),

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C) \quad (84)$$

$$= \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + r \partial_{xx}(\tilde{D}_2 C) + r^{-1} \partial_{yy}(\tilde{D}_2 C) + 2 \partial_{xy}(\tilde{D}_2 C). \quad (85)$$

$$\tilde{D}_2 = \frac{D_{xy}}{2} = \frac{d_2}{2}. \quad (86)$$

$$\tilde{A}_x = D_{xx} - r \tilde{D}_2 = \frac{d_1 + d_2}{2} - r \cdot \frac{d_2}{2} = \frac{d_1 + d_2(1 - r)}{2}. \quad (87)$$

$$\tilde{A}_y = D_{yy} - \frac{\tilde{D}_2}{r} = \frac{d_2}{2} - \frac{d_2/2}{r} = \frac{d_2}{2} \left(1 - \frac{1}{r} \right) = \frac{d_2(r - 1)}{2r}. \quad (88)$$

6.4 Non-negativity and mesh-aspect-ratio constraint

To have a unique solution, we need again positive operators (same reasoning as earlier).

Condition on $\tilde{D}_2$.

$$\tilde{D}_2 = \frac{d_2}{2} \geq 0 \iff d_2 \geq 0. \quad (89)$$

This holds by definition (77) for all (x, y, t) .

Condition on $\tilde{A}_y$.

$$\tilde{A}_y = \frac{d_2(r - 1)}{2r} \geq 0 \iff r - 1 \geq 0 \iff \boxed{r \geq 1, \quad \text{i.e.} \quad \Delta x \geq \Delta y.} \quad (90)$$

Condition on $\tilde{A}_x$. We now assume $r \geq 1$

$$\tilde{A}_x = \frac{d_1 + d_2(1-r)}{2} \geq 0 \iff d_1 \geq d_2(r-1). \quad (91)$$

But d_1 and d_2 depend on x , y , and t , which makes this one a bit trickier:

$$r - 1 \leq \inf_{x,y,t} \frac{d_1(x,y,t)}{d_2(x,y,t)}. \quad (92)$$

The way d_1 , d_2 , and β are defined, this leaves us with $d_1/d_2 \approx \epsilon^{-1}$. **We thus have to be careful how we set ϵ . Large values force us to have $r \approx 1$**

So we are left with:

$$1 \leq r \leq 1 + \rho, \quad r := \frac{\Delta x}{\Delta y}, \quad \rho := \inf_{x,y,t} \frac{d_1(x,y,t)}{d_2(x,y,t)}. \quad (93)$$

7 Programming this new method

I think much would change on the programming level. I think we can reuse what I did for the other method, but we need new initializations for our D -values and F -values now. We also must follow our new constraint on r , as outlined before.

I have not yet thought If this affects the CFL condition somehow... will need to do this next week combined with the source term formulation.

8 Source term formulation

The field (76) become time-dependent:

$$d_1(x, t) = \beta_{(1,0)}(x) C_{(1,0)}(t) + \alpha_{(1,0)}(x). \quad (94)$$

$\tilde{A}_x$, $\tilde{D}_2$ become time-dependent, thus all our factors must be recalculated each time-step. Maybe we can avoid recomputing the entire diffusion matrix by just modifying the right factors in our matrix. I have not yet further thought about this.

9 Questions

- Are there specific interesting testcases to think about?
- Should I do roundoff correction (like in Chap. 5.3) or leave the very small machine-precision drifts in the method?

References

- [1] M. Fey, “Multidimensional upwinding. part i. the method of transport for solving the euler equations,” *Journal of Computational Physics*, vol. 143, no. 1, pp. 159–180, 1998, ISSN: 0021-9991. DOI: <https://doi.org/10.1006/jcph.1998.5958>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0021999198959587>.